

Solution Architecture & Component Descriptions

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Component Description details

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides visual interfaces for onboarding, CRUD loan listings, charts, AI recommends, chat feed, and exports console.	React, Tailwind CSS, Recharts, Framer Motion, React Router
API Gateway / Middleware	Handles CORS headers, forwards JSON requests, and manages Nginx routing configurations.	Nginx default.conf (frontend Dockerfile) / FastAPI CORSMiddleware
Application / Logic Services	Authenticates credentials (PBKDF2), processes DTI calculations, connects to Gemini, and streams reports.	FastAPI, PyJWT, ReportLab, OpenPyXL, Gemini API wrapper
Data / Storage Layer	Persists tables for users, loans, recommendations, letter drafts, chat logs, and audits.	SQLite file database, SQLAlchemy ORM engine mappings